IB Mathematics Application and Interpretation HL IA

Investigation into the physical behavior of the one-degree-of-freedom suspension system of a quarter car model with road disturbance

Research Question: How can a second-order linear differential equation be mathematically transformed into a coupled first-order system to visualize the dynamic behavior of under-damped, critically damped, and over-damped automotive suspensions using phase portraits?

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I. Introduction

1.1 Rationale

This exploration started from my own experience of travelling by car and by bus. Going over the same kind of speed bump, the two vehicles felt completely different. For the car, I felt a stiff and harsh jolt, while the bus presented me with a lot of bouncing up and down after the bump, and this repeated bouncing made me feel dizzy. I wanted to understand why two vehicles respond so differently to the same disturbance, and whether mathematics could describe the difference.

A car's suspension has to balance two goals that work against each other. It has to keep the passengers comfortable, but it also has to keep the tires pressed onto the road so that the car can grip it. When a car goes over a speed bump or a rough road, energy from the bump is passed into the car body. If there was nothing to take this energy away, the car would keep shaking, which could damage the car and make the ride very uncomfortable.

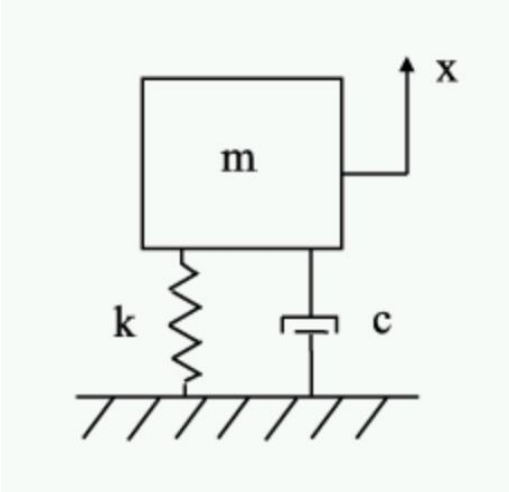


Figure1: One-degree-of-freedom quarter-car model (*1-DOF (Degree-of-freedom) Quarter Car Model.*, n.d.)

In order to analyze this mathematically, engineers often use the "Quarter Car Model". Instead of the whole car, this model only looks at one wheel and the quarter of the car body that sits above it. In this exploration, the simplest version, the one-degree-of-freedom (1-DOF) model, was used (Figure1). "One degree of freedom" means that only one number is needed to describe the motion: the height x of the car body above its resting position. The car body mass (m) is joined to the ground by a spring (k) and a damper (c) placed side by side, and it can only move up and down (*1-DOF (Degree-of-freedom) Quarter Car Model.*, n.d.).

The two parts that control the up-and-down motion after a bump are the spring and the damper (shock absorber), which takes energy away through fluid friction. How this system behaves depends on a quantity with no units called the damping ratio, written as zeta (ζ) (*Damping Ratio & Coefficient | Formula, Units & Examples - Lesson | Study.com*, n.d.)

$$\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{km}}$$

Where,

- c is the damping coefficient (N s m^{-1}), which shows how strongly the damper resists the motion.
- c_c is the critical damping coefficient, the value of c at which the system returns to its equilibrium as quickly as possible without oscillating.
- m is the mass of the system (kg)
- k is the spring constant (N m^{-1})

While a mathematically critically damped ($\zeta = 1$) or over-damped ($\zeta > 1$) system will return the car to its equilibrium ride height without any extra bouncing, these configurations are generally avoided in everyday passenger vehicles. This is because they make the ride stiff and harsh, as more of the force from the road is passed on to the car body. Instead, most passenger cars use an under-damped ratio between 0.2 and 0.4, so that the ride is softer and more comfortable (*Table 3 Vehicle Typical Damping Coefficient*, n.d.). This suggested that the difference I felt between the car and the bus could be related to how strongly each suspension is damped, so the damping coefficient became the variable of this exploration.

Traditionally, the vertical motion of this system is modeled using a single second-order differential equation. However, a single second-order equation describes only how displacement changes with time, and it does not show directly how displacement and velocity change together. By transforming this second-order equation into a system of coupled first-order differential equations (displacement and velocity), we are able to create a phase portrait. A phase portrait plots velocity against displacement, so it shows the position and the speed of the car body at the same time, and the whole motion for each damping scenario can be seen in one graph. As an aspiring engineer, I was eager to understand how suspensions deal with bumps, so that one day I could design vehicles that ride comfortably over different terrains.

1.2 Research question

How can a second-order linear differential equation be mathematically transformed into a coupled first-order system to visualize the dynamic behavior of under-damped, critically damped, and over-damped automotive suspensions using phase portraits?

1.3 Aims and Objectives

This exploration aims to:

1. Derive a mathematical model for a vehicle's suspension using Newtonian mechanics.

2. Transform the second-order differential equation into a coupled linear system using matrix algebra.
3. Use the characteristic equation of the matrix to find the damping coefficient at which the suspension becomes critically damped.
4. Solve the model exactly for a critically damped, overdamped, and underdamped suspension scenario using eigenvalues and eigenvectors.
5. Visualize the dynamic stability of the vehicle using phase portraits.
6. Evaluate the physical limitations of the mathematical model in the context of real-world automotive engineering.

II. System Parameters and Assumptions

For this investigation, we will model a single wheel of a vehicle using a simplified 1-DOF Quarter Car Model framework. To keep the mathematical analysis within the scope of matrix methods, we will focus exclusively on the displacement of the sprung mass after a bump has pushed it away from equilibrium and it is released, assuming the tire is rigid.

The system parameters are defined as follows:

- Mass of the car quarter (m): 400 kg. This represents the average mass of a mid-sized 1600 kg sedan divided equally across four wheels (Pramisha, 2025).
- Spring constant (k): 10000 N m^{-1}. This indicates how much force the spring applies per metre of compression. Together with m, it gives a natural angular frequency $\omega_n = \sqrt{\frac{k}{m}} = 5 rad\, s^{-1}$ ($f_n \approx 0.8\, Hz$), close to the body-bounce frequency of about 1 Hz typical of passenger cars (Routley, 2023).
- Damping coefficient of the shock absorber (c): This is the variable to be evaluated, measured in N s m^{-1}.
- Initial vertical displacement (x_0): 0.1 m. The body starts 10 cm above its equilibrium position, as if a 10cm bump had just pushed it up.
- Initial vertical velocity (v_0): 0 m s^{-1}. The body is momentarily at rest at this highest point.

In this investigation, several key assumptions were made to simplify the complex mechanical behaviour of a vehicle suspension into a solvable mathematical model.

Assumption	Explanation
Linearity of components	The model assumes that both the primary suspension spring and the damper behave linearly to be able to solve analytically with coupled differential equations.
1-DOF Quarter Car Model	The vehicle is treated as a 1-degree-of-freedom system, focusing

	only on the vertical displacement of the supported mass.
Rigid tire	The model assumes the tire is completely rigid, transferring any road disturbance directly to the suspension. This removes an additional spring element, allowing the suspension to be represented by a second-order differential equation rather than a fourth-order system.
Release from an initial displacement	The main analysis follows the body after it is released from rest at x=0.1 m, with no further road disturbance.
Displacement measured from static equilibrium	At equilibrium, the spring is already compressed by the car's weight, so the weight and the static spring force cancel. Measuring x from this position removes gravity from the equation.

Table1: Assumptions

III. Derivation of the second-order differential equation

To derive the second-order differential equation using Newton's second law, we look at the forces acting on the car's body as it moves vertically.

Newton's second law of motion states that the sum of all forces acting on an object is equal to its mass multiplied by its acceleration

$$\sum \boldsymbol{F = ma}$$

Acceleration is the second derivative of position (x) with respect to time (t). Therefore:

$$\sum \boldsymbol{F = m\frac{d^2x}{dt^2}}$$

When the car hits a bump and is displaced from its equilibrium position, two primary vertical forces are acting on the car body.

1. The spring force (F_s): According to Hooke's Law, a spring exerts a restoring force that is proportional to its displacement. The negative sign indicates that the force always acts in the opposite direction of the displacement.

$$\boldsymbol{F_s = -kx}$$

2. The damping force (F_d): The shock absorber creates viscous (fluid) friction. Unlike a spring, a damper's force depends on how fast the suspension is moving. Thus, this force is proportional to velocity (v), which is the first derivative of position ($\frac{dx}{dt}$). The negative sign indicates that the force always acts in the opposite direction of the velocity.

$$\boldsymbol{F_d = -cv = -c\frac{dx}{dt}}$$

We add all forces present into Newton's second law equation:

$$\boldsymbol{m\frac{d^2x}{dt^2} = F_s + F_d = -kx - c\frac{dx}{dt}}$$

By rearranging the terms to bring all variables to one side, we obtain a second-order linear differential equation:

$$\boldsymbol{m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0}$$

This is a linear second-order differential equation with constant coefficients. It shows that after a bump, the spring force pulls the car body back towards equilibrium while the damper force slows the body down, and together they bring the car back to rest.

IV. State-space form and eigenvalue analysis

A state-space equation turns a higher-order differential equation into a set of first-order differential equations that are linked together, and writes them using a matrix.

4.1 Transformation to a coupled system

In order to use matrices and eigenvalues, we introduce the velocity $v = \frac{dx}{dt}$ as a second variable. By isolating the acceleration term, we convert the single second-order equation into two coupled first-order equations:

$$\boldsymbol{a = -\frac{k}{m}x - \frac{c}{m}v}$$

$$\boldsymbol{\frac{dx}{dt} = v}$$

$$\boldsymbol{\frac{dv}{dt} = -\frac{k}{m}x - \frac{c}{m}v}$$

The two equations are coupled first-order equations as they are a system of simultaneous differential equations where the rate of change of each variable depends on the other variables in the system.

4.2 Matrix representation

In general, the system can be written as X'=MX, where

$$\boldsymbol{X = \begin{bmatrix} x \\ v \end{bmatrix}}$$

$$\boldsymbol{M = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}}$$

Substituting our specific constant parameters (m=400 and k = 10000), we can evaluate the coefficients:

$$\frac{k}{m} = \frac{10000}{400} = 25$$

The coupled system is now:

$$\frac{dx}{dt} = v$$

$$\frac{dv}{dt} = -25x - \frac{c}{400}v$$

This coupled system can be written in matrix form:

$$\mathbf{X'=MX}$$

$$\begin{bmatrix} x' \\ v' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -25 & -\frac{c}{400} \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

Since the derivative of $e^{\lambda t}$ is just $\lambda e^{\lambda t}$, an exponential function keeps the same shape when it is differentiated. This makes it a good guess for the solution of this system, so a solution of this form was tried:

$$\mathbf{X=P}e^{\lambda t}$$

Where P is a constant non-zero vector. Then we take the derivative of the vector with respect to time to find X'.

$$\mathbf{X'=}\lambda\mathbf{P}e^{\lambda t}$$

We plug both X and X' back into the original matrix equation (X'=MX):

$$\lambda\mathbf{P}e^{\lambda t}\mathbf{=M(P}e^{\lambda t}\mathbf{)}$$

Since $e^{\lambda t}$ is just a scalar number and never equals zero, we can divide both sides by it to remove it:

$$\lambda\mathbf{P=MP}$$

By rearranging and using the identity matrix(I), the equation is prepared to find its eigenvalues.

$$\mathbf{(M-}\lambda\mathbf{I)P=0}$$

4.3 The characteristic equation

If (M-λI) were invertible, the only solution would be P=0, which gives no motion. A non-zero eigenvector P therefore exists only when (M-λI) is non-invertible, that is, when its determinant is zero. We determine the eigenvalues by solving the characteristic equation.

$$\mathbf{det(M-}\lambda\mathbf{I)=0}$$

Substituting our matrix:

$$\begin{bmatrix} 0 & 1 \\ -25 & -\frac{c}{400} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$det\begin{bmatrix} -\lambda & 1 \\ -25 & -\frac{c}{400} - \lambda \end{bmatrix} = 0$$

Calculating the determinant:

$$(-\lambda)(-\frac{c}{400}-\lambda)-(1)(-25)=0$$

$$\lambda^2+(\frac{c}{400})\lambda+25=0$$

The behaviour of the suspension depends on the solutions of this quadratic equation. An advantage of finding the eigenvalues with a matrix is that the eigenvectors can also be found, and these appear in the phase portrait as the straight-line directions that trajectories follow.

4.4 The Discriminant, Eigenvalues, and System Stability

Because the characteristic equation is a quadratic polynomial, the eigenvalues depend on the discriminant.

$$\Delta =(\frac{c}{400})^2-4(1)(25)$$

$$\Delta =(\frac{c}{400})^2-100$$

Before looking at the three cases, the sum and product of the roots of $\lambda^2+(\frac{c}{400})\lambda+25=0$ show that the suspension is stable for every c>0:

$$\lambda_1+\lambda_2=-\frac{c}{400}<0, \lambda_1\lambda_2=25>0$$

If the eigenvalues are real, their product is positive, so they have the same sign, and their sum is negative, so both are negative. If they are complex, $\lambda = a \pm bi$ with 2a= $-\frac{c}{400}$, so the real part $a=-\frac{c}{800}$ is negative. Either way $e^{\lambda t}$ decays, so the car always returns to x=0. Only with no damper (c=0) would the eigenvalues be $\pm 5i$, giving an oscillation that never decays.

The sign of the discriminant decides which type of eigenvalues the system has, and therefore what kind of function describes the car's movement. The real part of an eigenvalue decides whether the motion dies out or grows, and the imaginary part decides whether the car oscillates:

Discriminant	Type of Eigenvalues	System Stability
Δ =0	One repeated real eigenvalue	If the discriminant is exactly zero, there is one repeated real eigenvalue. This is the boundary between real and complex eigenvalues. It is the critically damped case (ζ =1), which gives the fastest return to equilibrium without oscillation.
Δ >0	Two real, negative Eigenvalues	If the discriminant is positive, there are two different real eigenvalues. Since there are no imaginary numbers, the solution is made only of exponential decay terms ($e^{\lambda t}$). There

		are no sine or cosine terms, so nothing makes the car oscillate, and the motion simply decays toward zero. This is an over-damped system (ζ >1), where the car returns to equilibrium slowly without bouncing.
Δ <0	Complex Eigenvalues	If the discriminant is negative, the eigenvalues are complex ($\lambda = a \pm bi$). Using Euler's formula ($e^{ix} = \cos x + i \sin x$), the imaginary part turns into sine and cosine terms in the solution. Since sine and cosine are periodic, the car oscillates. This is an under-damped system (ζ <1), where the car bounces up and down with a decreasing amplitude, because a<0.

Table2: Discriminant, types of eigenvalues, and their system stability

V. Critically damped scenario

5.1 Damping coefficient for critically damped scenario

To find the exact damper setting required for critical damping, we set the discriminant to zero and solve for c:

$$\left(\frac{c}{400}\right)^2 - 100 = 0$$

$$\left(\frac{c}{400}\right)^2 = 100$$

$$\frac{c}{400} = \pm 10$$

But the damping coefficient must be positive:

$$c = 4000\ N\ s\ m^{-1}$$

Therefore, the shock absorber must be tuned to exactly $4000N\ s\ m^{-1}$ to achieve mathematical critical damping. Any value below this will cause the vehicle to oscillate, and any value above this will cause the vehicle to return to the equilibrium point without any oscillation.

5.2 Eigenvector and solution for critically damped scenario

At the critical damping threshold, $c = 4000N\ s\ m^{-1}$, the characteristic equation becomes $\lambda^2 + 10\lambda + 25 = 0$. This yields a single real eigenvalue.

$$(\lambda + 5)^2 = 0$$

$$\lambda = -5$$

The eigenvector satisfies (M+5I)P=0:

$$\begin{bmatrix} \mathbf{5} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-5} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

5x+v=0

v=-5x

$$\mathbf{P}=\begin{bmatrix} \mathbf{1} \\ \mathbf{-5} \end{bmatrix}$$

Both rows give the same equation, so there is only one independent eigenvector. The formula $X = C_1 e^{\lambda_1 t} P_1 + C_2 e^{\lambda_2 t} P_2$ therefore cannot be used, because it needs two different eigenvectors. Instead, the form $x(t) = (C_1 + C_2 t)e^{-5t}$ was tested (*ODE-Project Repeated Eigenvalues*, n.d.). Differentiating with the product rule:

$$\boldsymbol{x'(t) = (C_2 - 5C_1 - 5C_2 t)e^{-5t}}$$

$$\boldsymbol{x''(t) = (-10C_2 + 25C_1 + 25C_2 t)e^{-5t}}$$

Substituting into $x'' + 10x' + 25x = 0$ (the equation of motion divided by m=400):

$$\boldsymbol{e^{-5t}[(-10C_2 + 25C_1 + 25C_2 t) + (10C_2 - 50C_1 - 50C_2 t) + (25C_1 + 25C_2 t)] = e^{-5t} \times 0 = 0}$$

So the form satisfies the equation for every C_1 and C_2. Since v=x', the state vector can be written

$$\boldsymbol{X(t) = (C_1 + C_2 t)e^{-5t}\begin{bmatrix} 1 \\ -5 \end{bmatrix} + C_2 e^{-5t}\begin{bmatrix} 0 \\ 1 \end{bmatrix},}$$

Which shows that for large t the state vector points along the eigenvector (1, -5).

To find constants C_1 and C_2, the initial conditions (x=0.1m, v=0 m s^{-1} at t=0) were applied:

$$\boldsymbol{0.1 = (C_1 + C_2 \cdot 0)e^0}$$

$$\boldsymbol{C_1 = 0.1}$$

Applying the initial velocity condition (v=0 at t=0) to x' above:

$$\boldsymbol{0 = -0.5 + C_2}$$

$$\boldsymbol{C_2 = 0.5}$$

Thus, the exact solution defining the critically damped scenario is:

$$\boldsymbol{x(t) = (0.1 + 0.5t)e^{-5t}}$$

$$\boldsymbol{v(t) = -2.5te^{-5t}}$$

VI. Over-damped scenario

For an over-damped scenario, $\zeta > 1$. Since $c_c = 4000\ N\ s\ m^{-1}$, this condition corresponds to c>4000 N s m^{-1}. We tune the damping coefficient above the critical damping coefficient to 5000 N s m^{-1} ($\zeta = 1.25$). Our coefficient matrix becomes:

$$\boldsymbol{M} = \begin{bmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-12.5} \end{bmatrix}$$

6.1 Eigenvalues for over-damped scenario

Substituting c=5000 into our characteristic equation yields:

$$\boldsymbol{\lambda^2 + 12.5\lambda + 25 = 0}$$

We solve for the eigenvalues using the quadratic formula:

$$\boldsymbol{\lambda = \frac{-12.5 \pm \sqrt{12.5^2 - 100}}{2}}$$

$$\boldsymbol{\lambda = \frac{-12.5 \pm \sqrt{56.25}}{2}}$$

$$\boldsymbol{\lambda = \frac{-12.5 \pm 7.5}{2}}$$

This gives two real, distinct, negative eigenvalues:

$$\boldsymbol{\lambda_1 = -2.5}$$

$$\boldsymbol{\lambda_2 = -10}$$

6.2 Eigenvectors for over-damped scenario

Next, we determine the corresponding eigenvectors by solving the matrix equation for each eigenvalue:

$$\textbf{(M-}\boldsymbol{\lambda}\textbf{I)P=0}$$

For $\lambda_1 = -2.5$:

$$\begin{bmatrix} \mathbf{0-(-2.5)} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-12.5-(-2.5)} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{2.5} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-10} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

This translates to the linear equation:

$$\boldsymbol{2.5x + v = 0}$$

$$\boldsymbol{v = -2.5x}$$

By setting x=1, we find the first eigenvector:

$$\boldsymbol{P_1 = \begin{bmatrix} 1 \\ -2.5 \end{bmatrix}}$$

For $\lambda_2 = -10$:

$$\begin{bmatrix} \mathbf{0-(-10)} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-12.5-(-10)} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{10} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-2.5} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

This translates into the linear equation:

$$\boldsymbol{10x + v = 0}$$

$$\boldsymbol{v = -10x}$$

By setting x=1, we find the second eigenvector:

$$\boldsymbol{P_2 = \begin{bmatrix} 1 \\ -10 \end{bmatrix}}$$

In fact, the first row of (M-λI)P=0 always reads $-\lambda x + v = 0$, so for this matrix every eigenvector has the form $(1, \lambda)$. So the velocity is λ times the displacement.

6.3 Solving the coupled system for over-damped scenario

If the matrix yields two distinct eigenvalues, they will have two corresponding linearly independent eigenvectors P_1, P_2. The general solution is as follows:

$$\boldsymbol{X = C_1 e^{\lambda_1 t} P_1 + C_2 e^{\lambda_2 t} P_2}$$

Substituting the specific eigenvalues and eigenvectors:

$$\begin{bmatrix} \boldsymbol{x(t)} \\ \boldsymbol{v(t)} \end{bmatrix} = \boldsymbol{C_1 e^{-2.5t}} \begin{bmatrix} \boldsymbol{1} \\ \boldsymbol{-2.5} \end{bmatrix} + \boldsymbol{C_2 e^{-10t}} \begin{bmatrix} \boldsymbol{1} \\ \boldsymbol{-10} \end{bmatrix}$$

To find the specific constants C_1 and C_2, we substitute our initial conditions at time zero (displacement x = 0.1m, velocity v = 0m s^{-1}):

$$\begin{bmatrix} \boldsymbol{0.1} \\ \boldsymbol{0} \end{bmatrix} = \boldsymbol{C_1 e^0} \begin{bmatrix} \boldsymbol{1} \\ \boldsymbol{-2.5} \end{bmatrix} + \boldsymbol{C_2 e^0} \begin{bmatrix} \boldsymbol{1} \\ \boldsymbol{-10} \end{bmatrix}$$

This generates a system of linear equations:

$$\textbf{Equation 1: } \boldsymbol{C_1 + C_2 = 0.1}$$

$$\textbf{Equation 2: } \boldsymbol{-2.5C_1 - 10C_2 = 0}$$

From Equation 1, we can isolate C_2:

$$\boldsymbol{C_2 = 0.1 - C_1}$$

Substituting this into Equation 2:

$$\boldsymbol{-2.5C_1 - 10(0.1 - C_1) = 0}$$

$$\boldsymbol{-2.5C_1 - 1 + 10C_1 = 0}$$

$$\boldsymbol{7.5C_1 = 1}$$

$$\boldsymbol{C_1 = \frac{1}{7.5} = \frac{2}{15}}$$

Since we know the value of C_1,C_2 is:

$$\boldsymbol{C_2 = 0.1 - \frac{2}{15}}$$

$$\boldsymbol{C_2 = -\frac{1}{30}}$$

Thus, the model defining the displacement and velocity over time is fully derived. Each coefficient of v(t) is the coefficient of x(t) multiplied by the second component of its eigenvector, for example $\frac{2}{15} \times (-2.5) = -\frac{1}{3}$:

$$\boldsymbol{x(t) = \frac{2}{15} e^{-2.5t} - \frac{1}{30} e^{-10t}}$$

$$\boldsymbol{v(t) = -\frac{1}{3} e^{-2.5t} + \frac{1}{3} e^{-10t}}$$

VII. Under-damped scenario

7.1 Eigenvalues for under-damped scenario

To optimize the vehicle for comfort, engineers target a damping ratio typically around 0.2 to 0.4. For this research, a damping ratio of $\zeta = 0.3$ was selected. Using the formula $\zeta = \frac{c}{c_c}$, we can find the

damping coefficient c:

$$\mathbf{0.3} = \frac{\boldsymbol{c}}{\mathbf{4000}}$$

$$\mathbf{c=1200N\ s\ } \boldsymbol{m^{-1}}$$

Using this damping ratio, the matrix becomes:

$$\boldsymbol{M} = \begin{bmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{-25} & \mathbf{-3} \end{bmatrix}$$

The characteristic equation is now $\lambda^2 + 3\lambda + 25 = 0$. Because the discriminant is negative ($\triangle = 9 - 100 = -91$), the quadratic formula yields complex eigenvalues:

$$\boldsymbol{\lambda} = \frac{\mathbf{-3 \pm \sqrt{-91}}}{\mathbf{2}} = \mathbf{-1.5 \pm \frac{\sqrt{91}}{2}}\boldsymbol{i}$$

That is, $\lambda \approx -1.5 \pm 4.77i$.

7.2 Solving the coupled system for under-damped scenario

The method is the same as for the over-damped scenario: find the eigenvectors, then combine the two solutions $Pe^{\lambda t}$. The only difference is that λ, and therefore P, are now complex numbers. To keep this working short, the eigenvalues are written as $\lambda = a \pm bi$, where a=-1.5 and b=$\frac{\sqrt{91}}{2}$.

The eigenvectors are found in the same way as Section 6.2, by solving $(M - \lambda I)P = 0$.

For $\lambda_1 = a \pm bi$:

$$\begin{bmatrix} \boldsymbol{-\lambda_1} & \mathbf{1} \\ \mathbf{-25} & \boldsymbol{-3-\lambda_1} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

The first row gives $-\lambda_1 x + v = 0, so\ v = \lambda_1 x$. By setting x=1, we find the first eigenvector:

$$\boldsymbol{P_1} = \begin{bmatrix} \mathbf{1} \\ \boldsymbol{\lambda_1} \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ \mathbf{-1.5 + \frac{\sqrt{91}}{2}}\boldsymbol{i} \end{bmatrix}$$

To check it, the second row gives $-25 + (-3 - \lambda_1)\lambda_1 = -(\lambda_1^2 + 3\lambda_1 + 25)$, which is 0 because λ_1 solves the characteristic equation. So both rows agree. This is the same pattern as in Section 6.2: the velocity is λ times the displacement, but now λ is complex. In the same way, $\lambda_2 = a - bi$ gives the second eigenvector:

$$\boldsymbol{P_2} = \begin{bmatrix} \mathbf{1} \\ \boldsymbol{\lambda_2} \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ \mathbf{-1.5 - \frac{\sqrt{91}}{2}}\boldsymbol{i} \end{bmatrix}$$

As in Section 6.3, the general solution is $X = Ae^{\lambda_1 t}P_1 + Be^{\lambda_2 t}P_2$, where A and B are constants.

Since the first component of both eigenvectors is 1, the displacement is

$$\boldsymbol{x(t) = Ae^{(a+bi)t} + Be^{(a-bi)t} = e^{at}(Ae^{ibt} + Be^{-ibt})}$$

Euler's formula gives $e^{ibt} = cos(bt) + isin(bt)$ and $e^{-ibt} = cos(bt) - isin(bt)$. Substituting these:

$$\boldsymbol{x(t) = e^{at}[(A+B)cos(bt) + i(A-B)sin(bt)]}$$

Let $C_1 = A + B$ and $C_2 = i(A - B)$. The displacement of a real car must be a real number at every time, so C_1 and C_2 must be real numbers, even though A and B themselves are complex. This gives the real general solution:

$$\boldsymbol{x(t) = e^{at}(C_1 cos(bt) + C_2 sin(bt))}$$

This also shows what each part of the eigenvalue does: the real part a=-1.5 gives the factor $e^{-1.5t}$, which makes the motion smaller over time, and the imaginary part $b = \frac{\sqrt{91}}{2}$ gives the cosine and sine terms, which make the car oscillate.

Substituting the eigenvalue:

$$\boldsymbol{x(t) = e^{-1.5t}(C_1 cos(\frac{\sqrt{91}}{2}t) + C_2 sin(\frac{\sqrt{91}}{2}t))}$$

To find the specific constants C_1 and C_2, we apply the initial condition for displacement (x = 0.1 m at t = 0):

$$\boldsymbol{0.1 = e^0(C_1 cos(0) + C_2 sin(0))}$$

$$\boldsymbol{C_1 = 0.1}$$

Next, the velocity function v(t) is derived by taking the first derivative of x(t) using the product rule:

$$\boldsymbol{v(t) = -1.5e^{-1.5t}(C_1 cos(\frac{\sqrt{91}}{2}t) + C_2 sin(\frac{\sqrt{91}}{2}t)) + e^{-1.5t}(-C_1\frac{\sqrt{91}}{2}sin(\frac{\sqrt{91}}{2}t) + C_2\frac{\sqrt{91}}{2}cos(\frac{\sqrt{91}}{2}t))}$$

Grouping the cosine and sine terms yields:

$$\boldsymbol{v(t) = e^{-1.5t}[(-1.5C_1 + \frac{\sqrt{91}}{2}C_2)cos(\frac{\sqrt{91}}{2}t) - (1.5C_2 + \frac{\sqrt{91}}{2}C_1)sin(\frac{\sqrt{91}}{2}t)]}$$

We apply the initial condition for velocity (v=0 m s^{-1} at t=0)

$$\boldsymbol{0 = e^0[(-1.5C_1 + \frac{\sqrt{91}}{2}C_2)cos(0) - (1.5C_2 + \frac{\sqrt{91}}{2}C_1)sin(0)]}$$

$$\boldsymbol{0 = -1.5C_1 + \frac{\sqrt{91}}{2}C_2}$$

Substituting $C_1 = 0.1$:

$$\boldsymbol{0.15 = \frac{\sqrt{91}}{2}C_2}$$

$$\boldsymbol{C_2 = \frac{0.3}{\sqrt{91}}}$$

Thus, the equations for the under-damped scenario are fully derived:

$$\boldsymbol{x(t) = e^{-1.5t}(0.1cos(\frac{\sqrt{91}}{2}t) + \frac{0.3}{\sqrt{91}}sin(\frac{\sqrt{91}}{2}t))}$$

When C_1 and C_2 are substituted back into the v(t) equation, the cosine term disappears, because its coefficient $-1.5C_1 + \frac{\sqrt{91}}{2}C_2$ is exactly the expression set to 0 above. The sine coefficient becomes $1.5C_2 + \frac{\sqrt{91}}{2}C_1 = 1.5 \times \frac{0.3}{\sqrt{91}} + 0.1 \times \frac{\sqrt{91}}{2} = \frac{0.45+4.55}{\sqrt{91}} = \frac{5}{\sqrt{91}}$, so the final velocity model is:

$$\boldsymbol{v(t) = -\frac{5\sqrt{91}}{91}e^{-1.5t}sin(\frac{\sqrt{91}}{2}t) \approx -0.524e^{-1.5t}sin(4.77t)}$$

The system eventually decays to 0, while the sine function causes the suspension to oscillate at an angular frequency of 4.77 rad s^{-1} before settling. One full oscillation takes $T = \frac{2\pi}{\frac{\sqrt{91}}{2}} \approx 1.32\ s$. In that time the factor $e^{-1.5t}$ is multiplied by $e^{-1.5T} \approx 0.139$, so after each full oscillation the size of the motion is only about 14% of what it was.

VIII. Phase portrait representation

A phase portrait is a graph of velocity (v) on the vertical axis against displacement (x) on the horizontal axis. It shows how the state of the car body changes, without plotting time itself. Each portrait below shows many trajectories (grey, with arrows in the direction of increasing t), the lines through the origin in the directions of the real eigenvectors (dashed and dotted), and, in blue, the trajectory of this investigation. By starting all three scenarios at the initial coordinate (0.1, 0), we can compare their trajectories toward the equilibrium origin (0,0). Because $\frac{dx}{dt} = v$, every trajectory moves to the right above the x-axis and to the left below it. Using Python (Matplotlib and NumPy libraries), the dynamic behaviours of the suspensions were visualized in a phase portrait.

8.1 Critically-damped scenario (c=4000N s m^{-1})

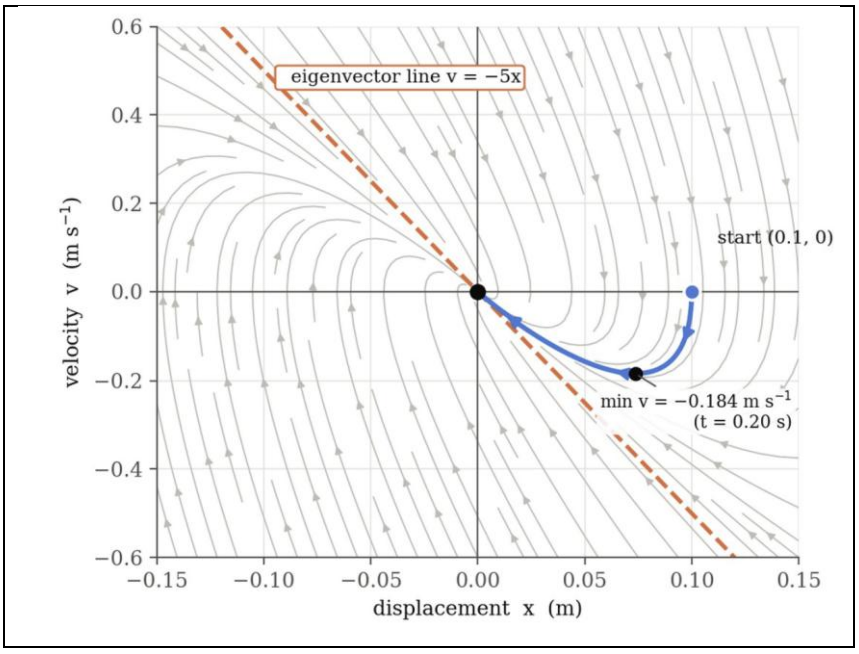


Figure2: Phase portrait of critically damped scenario

$$\boldsymbol{x(t) = (0.1 + 0.5t)e^{-5t}}$$

$$\boldsymbol{v(t) = -2.5te^{-5t}}$$

The phase portrait exhibits a degenerate stable node at the origin (0, 0). Mathematically, because the system has a single repeated real eigenvalue λ=-5, the solution takes the form shown above. The graph displays the trajectory approaching the origin along the only eigenvector line, v=-5x: from the solution, $\frac{v}{x} = \frac{-2.5t}{0.1+0.5t} \to -5$ as t→ ∞. Every other trajectory in Figure 2 also enters the origin tangent to this line. The body reaches its greatest downward speed, 0.184 m $s^{-1}, at\ t = 0.2\ s$ (where $\frac{dv}{dt} = -2.5e^{-5t}(1 - 5t) = 0$), while x is still 0.074m. The blue trajectory never crosses the v-axis, so the body never overshoots equilibrium.

8.2 Over-damped scenario (c=5000N s m^{-1})

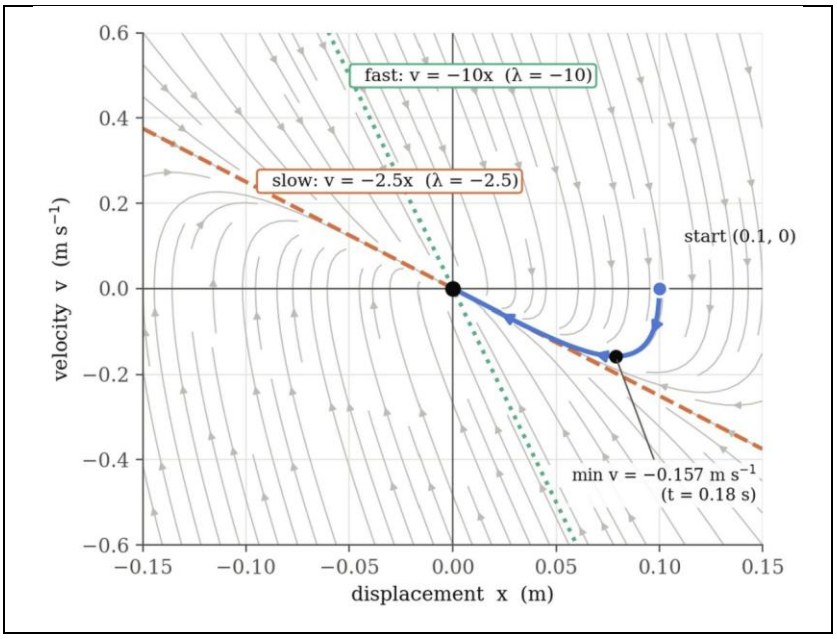


Figure3: Phase portrait of over-damped scenario

$$\boldsymbol{x(t) = \frac{2}{15}e^{-2.5t} - \frac{1}{30}e^{-10t}}$$

$$\boldsymbol{v(t) = -\frac{1}{3}e^{-2.5t} + \frac{1}{3}e^{-10t}}$$

The portrait shows a stable node characterized by two distinct real eigenvalues, λ_1=-2.5 and λ_2=-10. The trajectory depends on these two rates of decay. It is initially influenced by both decay rates, but as t→ ∞, the term associated with λ_2=-10 vanishes quickly, so the trajectory lines up with the eigenvector of the slower eigenvalue λ_1=-2.5 ($\frac{v}{x} \to -2.5$). Dividing the numerator and denominator of $\frac{v}{x}$ by $e^{-2.5t}$

gives $\frac{v}{x} = \frac{-\frac{1}{3}+\frac{1}{3}e^{-7.5t}}{\frac{2}{15}-\frac{1}{30}e^{-7.5t}} \to \frac{-\frac{1}{3}}{\frac{2}{15}} = -2.5$ as $t \to \infty$. It might be expected that the more negative eigenvalue $\lambda_2 = -10$ would control the motion, but the opposite occurs. Its term decays faster, so it disappears first, leaving the less negative eigenvalue to control the final approach.

Trajectories that start far from the origin first run parallel to the fast eigenvector line v=-10x and then turn onto the slow line v=-2.5x. Comparing Figures 2 and 3 shows why the over-damped suspension is more sluggish: its greatest downward speed is only 0.157 m s^{-1} (at t=$\frac{ln\,4}{7.5} \approx 0.18s$), compared with 0.184 m s^{-1}, because the stronger damper resists the motion. Most of its return happens along the slow eigenvector, whose term decays like $e^{-2.5t}$, compared with e^{-5t} in the critical scenario.

8.3 Under-damped scenario (c = 1200 N s m^{-1})

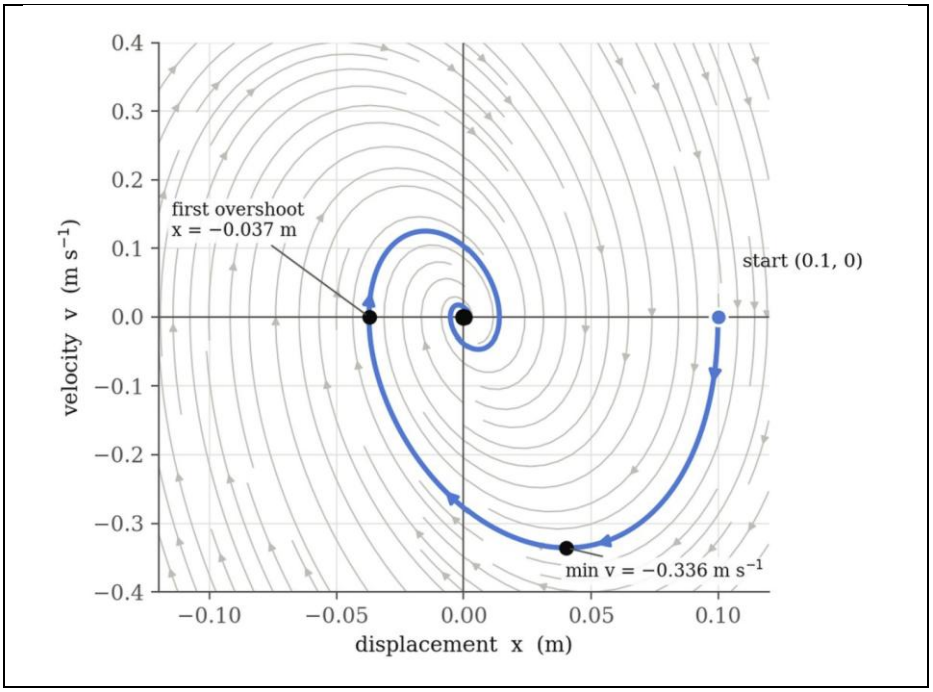


Figure4: Phase portrait of under-damped scenario

$$\boldsymbol{x(t) = e^{-1.5t}(0.1cos(\frac{\sqrt{91}}{2}t) + \frac{0.3}{\sqrt{91}}sin(\frac{\sqrt{91}}{2}t))}$$

$$\boldsymbol{v(t) = -\frac{5\sqrt{91}}{91}e^{-1.5t}sin(\frac{\sqrt{91}}{2}t)}$$

The portrait displays an asymptotically stable spiral at the origin, which comes from the complex eigenvalues $\lambda = -1.5 \pm \frac{\sqrt{91}}{2}i$. There are no real eigenvectors, so there are no straight-line trajectories. As shown in Section 7.2, the real part -1.5 makes the motion smaller over time, so the trajectory moves closer to the origin, and the imaginary part $\frac{\sqrt{91}}{2}$ makes the car oscillate, so the trajectory goes around the origin. Together these make the spiral. The direction of rotation follows from the equations: at the

starting point (0.1, 0), $\frac{dv}{dt} = -25(0.1) - 3(0) = -2.5 < 0$, so the state moves downward from the positive x-axis and the spiral is clockwise. Each loop of the spiral is one bounce of the car. Each time the trajectory crosses the x-axis, v=0, so the body stops for a moment and changes direction.

From v(t), v=0 when $sin(\frac{\sqrt{91}}{2}t) = 0$, which happens when $\frac{\sqrt{91}}{2}t = k\pi$:

$$\boldsymbol{t = \frac{k\pi}{\frac{\sqrt{91}}{2}}, k = 1, 2, 3, \ldots}$$

At these times the sine term in x(t) is 0 and cos(kπ)=(-1)^k, so substituting them into x(t) gives

$$\boldsymbol{x = 0.1(-1)^k e^{-1.5t}}$$

k	Time t (s)	Displacement x (cm)	What the car body does
1	0.66	-3.7	Lowest point, starts moving up
2	1.32	+1.4	Highest point, starts moving down
3	1.98	-0.5	Lowest point, starts moving up
4	2.63	+0.2	Highest point, starts moving down

Table3: Points where the under-damped car body changes direction

Each value is about 37% of the one before it, because half an oscillation multiplies the size of motion by $e^{-1.5(\frac{T}{2})} \approx 0.37$. After two crossings, which is one full oscillation, only $0.37^2 \approx 0.14$ is left, which matches the 14% found in Section 7.2. This means the car body changes direction three times by more than 2 mm before the bouncing becomes too small to notice. This repeated up-and-down movement is what made me dizzy on the bus.

IX. Conclusion

The investigation answered the research question "How can a second-order linear differential equation be mathematically transformed into a coupled first-order system to visualize the dynamic behavior of under-damped, critically damped, and over-damped automotive suspensions using phase portraits?" Converting the second-order differential equation into a coupled first-order system made it possible to draw its solution as phase portraits. The characteristic equation of the matrix turned out to be the same quadratic that comes directly from the second-order equation, so both approaches give the same eigenvalues. The advantage of the matrix form is that it gives both displacement and velocity together, and it treats them as one state vector. It also gives eigenvectors, which are used in the phase portraits as the straight lines that trajectories approach (v=-5x and v=-2.5x). The sign of the discriminant decides the type of eigenvalue, and the sum and product of the eigenvalues show that the suspension is stable for every c>0.

Scenario	Damping Coefficient(c)	Eigenvalues	System stability
Critically Damped	4000 N s m^{-1}	-5 (repeated)	Degenerate stable node
Over-Damped	5000 N s m^{-1}	-2.5, -10	Stable node
Under-Damped	1200 N s m^{-1}	$-1.5 \pm \frac{\sqrt{91}}{2} i$	Asymptotically stable spiral (clockwise)

Table4: The three scenarios and their eigenvalues and system stability

From an engineering perspective, the phase portraits show the comfort-handling trade-off in vehicle movements. An over-damped stable node prevents the car from bouncing, but it returns more slowly than the critically damped one, because most of its motion follows the slow eigenvector v=-2.5x. The under-damped stable spiral bounces the passenger up and down three times before the motion becomes smaller than 2mm (Table3). The critically damped suspension is the fastest to settle without any overshoot. The answer to "which damping is the best" depends on what is being optimized: settling time, overshoot or comfort.

This also answers the question that started the exploration. The repeated bouncing I felt on the bus corresponds to complex eigenvalues and a spiral in the phase plane, while a suspension with real eigenvalues returns to equilibrium without crossing it. However, the harsh jolt I felt in the car could not be explained by this model. In the model, the car body starts at rest at x=0.1 m, so at the start the damper does not push on it at all. In a real bump, the road suddenly pushes the bottom of the damper upwards, and a stiffer damper passes more of this push on to the car body, which is felt as a jolt. To show this, the movement of the road would have to be added to the model.

X. Evaluation

This exploration successfully mapped the second-order differential equation into phase portraits. However, the investigation has several assumptions that limit its accuracy when applied to real-world engineering.

Limitations	Explanation	Improvements
Linearity	The differential equations assumed that both the primary	Non-linear equations usually have no exact solution, so future investigation should

	spring and the damper behave linearly. In reality, the primary suspension becomes stiffer near full compression (bump stops), and the tyre's stiffness is also non-linear.	employ a step-by-step numerical method, such as Euler's method, to approximate the vehicle's state step-by-step. This would allow for a more accurate representation of the phase portrait under real-world road conditions.
1-DOF simplification	The investigation utilized a 1-DOF quarter-car model, treating the pneumatic tire as a completely rigid boundary. In reality, the tire acts as an additional spring and possesses an unsprung mass. This means a 2-DOF system is more accurate for representing reality.	Incorporating the unsprung mass and the tire's spring rate would expand the system to four state variables, resulting in a 4x4 matrix. While this requires computational software to extract the eigenvalues from the quartic (degree 4) characteristic equation, it would allow for a highly accurate phase portrait that maps both the body of the vehicle and tire compression simultaneously.
Environmental Dynamics	The model assumes constant system parameters (m, k, c). In reality, shock absorber performance is highly sensitive to thermal fluctuations. As the damper fluid heats up during repeated compression cycles, its viscosity changes, causing the damping coefficient to drop. This is a phenomenon known as shock fade.	Future research should incorporate a temperature-dependent variable for the damping coefficient. By modeling c(T), where T is a function of time and energy dissipation, the phase portrait could account for performance degradation during long driving conditions.
Idealized road disturbance	The main model starts from rest at x=0.1 m. In reality, a real bump is a ramp, not a step.	Model the road profile as a ramp or a half-sine bump.

Table5: Limitations and improvements

XI. Extension

To show the jolt I felt in the car, a future investigation could model the car actually driving over the

bump, instead of starting the car body at its highest point, 0.1 m above its resting position, as in this exploration. As the wheel goes over the bump, the road surface under it rises and pushes the bottom of the spring and the damper upwards. If y is the height of the road surface under the wheel, the spring and the damper would react to the difference between the movement of the car body and the road, so Newton's second law would become $m\frac{d^2x}{dt^2} = -k(x-y) - c(\frac{dx}{dt} - \frac{dy}{dt})$ (Dr Arnaldo Delli Carri, Coventry University, n.d.). The road could be modelled as a sudden step, a ramp, or a half-sine bump. Because the damper term depends on how fast the road moves, I would expect a stiffer damper to pass more of the bump onto the car body, giving a harder jolt. This could be tested by finding the largest acceleration of the car body for the three damping coefficients used in this exploration, either with the same eigenvalue method (using new initial conditions after a step) or with Euler's method for a ramp or a half-sine bump. If the over-damped suspension gives the largest acceleration, it would explain why the car felt harsh while the bus felt softer but bouncier.

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